4.3 Types of Controllers in Laravel

Laravel supports several controller types to support various development patterns:

4.3.1 Basic Controllers

Standard classes with multiple methods for different endpoints.

4.3.2 Resource Controllers

Ideal for CRUD operations using RESTful conventions.

```
php artisan make:controller ProductController --resource
```

This creates predefined methods like: - `index()` - `create()` - `store()` - `show()` - `edit()` - `update()` - `destroy()`

Register using:

```
Route::resource('products', ProductController::class);
```

4.3.3 Invokable Controllers

Useful for single-action endpoints.

```
php artisan make:controller ContactController --invokable
```

Define a route:

```
Route::get('/contact', ContactController::class);
```

4.4 Grouping Routes with Controllers

You can group routes for better organization:

```
Route::controller(UserController::class)->group(function () {
    Route::get('/users', 'index');
    Route::get('/users/{id}', 'show');
});
```

4.5 Request Injection and Dependency Injection